Worksheet 6.2 — Vector Operations (Geometric)

Grade 12 Calculus and Vectors (MCV4U) · Unit 6: Geometry and Algebra of Vectors

The Big Ideas

Add tip-to-tail (triangle law); subtract via $\vec{a} - \vec{b} = \vec{a} + (-\vec{b})$; scale by k to change length (and flip if $k < 0$).

- Addition: tip-to-tail; resultant from first tail to last tip.
- Subtraction: $\vec{a} - \vec{b} = \vec{a} + (-\vec{b})$.
- $k\vec{v}$ has length $|k||\vec{v}|$; reversed if $k < 0$.

The Big Picture

Adding vectors

Add vectors **tip-to-tail**: the resultant runs from the first tail to the last tip. The parallelogram rule gives the same result.

Scalar multiples

Multiplying by a scalar scales the length and, if negative, reverses the direction while keeping the line of action.

Subtraction

Subtracting a vector means adding its opposite, which the triangle of vectors makes easy to picture.

Worked Examples

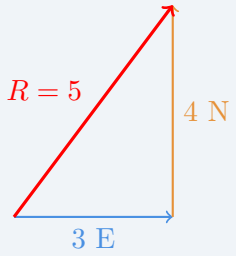
Example 1 — Perpendicular resultant

Add 3 east and 4 north; find the resultant.

Solution. **Step 1 (the two vectors are perpendicular, so use the Pythagorean theorem for magnitude):** $|\vec{R}| = \sqrt{3^2 + 4^2} = \sqrt{9 + 16} = \sqrt{25}$.

Step 2 (find the direction with $\tan^{-1}$ of opposite over adjacent): $\theta = \tan^{-1}\frac{4}{3} \approx 53°$ north of east.

Conclusion: magnitude 5, direction $\approx 53°$ N of E:



Example 2 — Triangle law

How do you draw $\vec{a}+\vec{b}$?

Solution. **Step 1 (draw $\vec{a}$ first):** start anywhere and draw $\vec{a}$.

Step 2 (attach $\vec{b}$ tip-to-tail): place the tail of $\vec{b}$ at the tip (arrowhead) of $\vec{a}$.

Step 3 (draw the resultant): the sum $\vec{a}+\vec{b}$ runs from $\vec{a}$'s starting tail to $\vec{b}$'s final tip.

Conclusion: tip-to-tail, then close the triangle from the first tail to the last tip.

Example 3 — Scalar multiple

Describe $3\vec{v}$ and $-2\vec{v}$.

Solution. **Step 1 (scalar multiplication scales the length by $|k|$):** for $3\vec{v}$, $|k|=3$; for $-2\vec{v}$, $|k|=2$.

Step 2 (the sign of k determines direction): positive k keeps the same direction; negative k reverses it.

Conclusion: $3\vec{v}$ is three times as long, same direction; $-2\vec{v}$ is twice as long, opposite direction.

Example 4 — Subtraction

How do you draw $\vec{a}-\vec{b}$?

Solution. **Step 1 (rewrite subtraction as addition of the opposite):** $\vec{a}-\vec{b}=\vec{a}+(-\vec{b})$.

Step 2 (draw $-\vec{b}$, i.e. $\vec{b}$ reversed): flip $\vec{b}$'s direction, keeping its length.

Step 3 (add tip-to-tail): attach $-\vec{b}$'s tail to $\vec{a}$'s tip, as in ordinary addition.

Conclusion: draw $\vec{a}$, then add the reversed $\vec{b}$ tip-to-tail.

Example 5 — Resultant force

Forces 6 N east and 8 N north. Find the resultant.

Solution. **Step 1 (perpendicular components, use Pythagorean theorem):** $|\vec{R}|=\sqrt{6^2+8^2}=\sqrt{36+64}=\sqrt{100}$.

Step 2 (find the direction): $\theta=\tan^{-1}\frac{8}{6}\approx 53^\circ$ north of east.

Conclusion: 10 N at $\approx 53^\circ$ N of E.

Example 6 — Perpendicular resultant

Resultant magnitude of 5 east and 12 north?

Solution. **Step 1 (Pythagorean theorem, since the components are perpendicular):** $\sqrt{5^2+12^2}=\sqrt{25+144}=\sqrt{169}$.

Conclusion: 13.

Example 7 — Scalar multiple

Describe $-\frac{1}{2}\vec{v}$.

Solution. **Step 1 (magnitude scales by $|k|=\frac{1}{2}$):** half the original length.

Step 2 (negative k reverses direction): points opposite to $\vec{v}$.

Conclusion: half the length, opposite direction.

Example 8 — Perpendicular resultant

Resultant magnitude of 9 east and 12 north?

Solution. **Step 1 (Pythagorean theorem):** $\sqrt{9^2+12^2}=\sqrt{81+144}=\sqrt{225}$.

Conclusion: 15.

Example 9 — Direction

Direction of the resultant of 3 east, 4 north?

Solution. **Step 1 (set up the tangent ratio, opposite over adjacent):** $\tan\theta=\frac{4}{3}$ (north component over east component).

Step 2 (solve for θ): $\theta=\tan^{-1}\frac{4}{3}\approx 53^\circ$.

Conclusion: $\approx 53^\circ$ north of east.

Your Turn — Show Your Work

For each question, show all steps clearly in the space provided.

Question 1

Resultant magnitude of 5 east and 12 north?

Question 2

Describe $4\vec{v}$.

Question 3

Describe $-\frac{1}{3}\vec{v}$.

Question 4

Resultant magnitude of 8 east and 15 north?

Question 5

What law adds vectors tip-to-tail?

Question 6

Resultant magnitude of 6 east and 8 north?

Question 7

Describe $-3\vec{v}$.

Question 8

Resultant magnitude of 7 east and 24 north?

Question 9

How do you express $\vec{a} - \vec{b}$ as an addition?

Question 10

Direction (N of E) of the resultant of 5 east, 5 north?

Question 11

Resultant magnitude of 9 east and 40 north?

Question 12

Length of $5\vec{v}$ if $|\vec{v}| = 3$?

Question 13 — Challenge

Two forces, 10 N east and 24 N north, act on an object. Find the magnitude and direction of the resultant.

Question Recap & Answer Key

Question Recap

1. Resultant magnitude of 5 east and 12 north?
2. Describe $4\vec{v}$.
3. Describe $-\frac{1}{3}\vec{v}$.
4. Resultant magnitude of 8 east and 15 north?
5. What law adds vectors tip-to-tail?
6. Resultant magnitude of 6 east and 8 north?
7. Describe $-3\vec{v}$.
8. Resultant magnitude of 7 east and 24 north?
9. How do you express $\vec{a} - \vec{b}$ as an addition?
10. Direction (N of E) of the resultant of 5 east, 5 north?
11. Resultant magnitude of 9 east and 40 north?
12. Length of $5\vec{v}$ if $|\vec{v}| = 3$?
13. Two forces, 10 N east and 24 N north, act on an object. Find the magnitude and direction of the resultant.

✓ Answer Key

1. 13
2. 4Œ length, same direction
3. $\frac{1}{3}$ length, opposite
4. 17
5. triangle law
6. 10
7. 3Œ length, opposite direction
8. 25
9. $\vec{a} + (-\vec{b})$
10. 45°
11. 41
12. 15
13. $\sqrt{100 + 576} = 26$ N at $\tan^{-1}\frac{24}{10} \approx 67°$ N of E